The Chinese University of Hong Kong, Shenzhen
School of Data Science

MAT3007 · Homework 5

Due: 11:59 pm, July 14, 2023

Instructions:

- Homework problems must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
- You must submit your assignment in Blackboard. If some problems need coding, you should paste your code in the file. Please upload only **one** file(pdf). The file name should be in the format **last name-first name-hw6**.
- The homework must be written in English.
- Late submission will not be graded.
- Each student **must not copy** homework solutions from another student or from any other source.

Problem 1 (20pts). Consider the following function:

$$f(x,y,z) = 3x^2 - 2x - 2xy + 3y^2 - 2y - 2zy + 3z^2 - 2z - 2xz$$

(a). Considering the 1st-order necessary condition, try to find the candidate minimizers of $f(x,y,z)$.

(b). Considering the 2nd-order sufficient condition, whether these candidates are indeed local minimizers?

(c). Is (0,0,0) a local minimizer? Why?

Solution:

1. By 1st-order necessary condition, $\nabla f(x^*) = 0$

$$\begin{aligned} 6x - 2y - 2z &= 2 \\ 6y - 2z - 2x &= 2 \\ 6z - 2x - 2y &= 2 \end{aligned}$$

   Solving the above equations, we get candidate minimizer (1,1,1).

2. We calculate the Hessian matrix:

$$H = \begin{pmatrix} 6 & -2 & -2 \\ -2 & 6 & -2 \\ -2 & -2 & 6 \end{pmatrix}$$

   Since all eigenvalues are positive, this matrix is positive definite. By 2nd-order sufficient condition, the candidate point is indeed a local minimum.

3. No. Since $\nabla f(0,0,0) = [-2;-2;-2]$, it can not be a local minimizer for unconstrained problem.

□

Problem 2 (20pts). Given a symmetric matrix $A \in R^{n\times n}$, consider the following problem:

$$\begin{array}{rl} \min_{x\in R^n} & x^T Ax \\ \text{subject to} & 2 - x^T x = 0 \end{array}$$

(a). Give the KKT conditions of this problem.

(b). If A is positive definite without repeated eigenvalues, how many different KKT points are there at most?

(c). If A is positive definite without repeated eigenvalues, what is the minimum value of this problem, and how many local minimizers? (Hint:According to Rayleigh quotient, $min\{x^T Ax/(x^T x)\} = \lambda_{min}$, where λ_{min} is the minimum eigenvalue of A)

Solution:

1. The KKT conditions are $Ax - \lambda x = 0$, $2 - x^T x = 0$, where $\lambda \in R$.
2. x is a KKT point of this problem if and only if x is a eigenvector of A, and the magnitude of the vector is $\sqrt{2}$. If A has no repeated eigenvectors, each eigenvector x_i corresponds to $\pm\sqrt{2}q_i$, where q_i is the normalized vector of x_i, then there are 2n KKT points at most.
3. According to Rayleigh quotient, $min\{x^T Ax/(x^T x)\} = \lambda_{min}$, where λ_{min} is the minimum eigenvalue of A, the minimum value of this problem is $2\lambda_{min}$, and there are 2 local minimizers.

□

Problem 3 (20pts). Construct the KKT conditions for the following linear program:

$$\begin{array}{rl} \text{maximize} & 3x_1 + x_2 + 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 \le 5 \\ & x_1 + 2x_2 + 2x_3 \le 8 \\ & x_1, x_2, x_3 \ge 0 \end{array}$$

Solution:

Let λi be the Lagrangian multiplier of the ith constraint.
Lagrangian Function:

$$L(x,\lambda) = -3x_1 - x_2 - 4x_3 + \lambda_1(x_1 + 3x_2 + x_3 - 5) + \lambda_2(x_1 + 2x_2 + 2x_3 - 8)$$

Main condition:

$$\begin{array}{l} -3 + \lambda_1 + \lambda_2 \ge 0 \\ -1 + 3\lambda_1 + 2\lambda_2 \ge 0 \\ -4 + \lambda_1 + 2\lambda_2 \ge 0 \end{array}$$

Complementary Condition:

$$\lambda_1(x_1 + 3x_2 + x_3 - 5) = 0$$
$$\lambda_2(x_1 + 2x_2 + 2x_3 - 8) = 0$$
$$x_1(-3 + \lambda_1 + \lambda_2) = 0$$
$$x_2(-1 + 3\lambda_1 + 2\lambda_2) = 0$$
$$x_3(-4 + \lambda_1 + 2\lambda_2) = 0$$

Dual Feasibility:

$$\lambda_1 \geq 0$$
$$\lambda_2 \geq 0$$

Primal Feasibility:

$$x_1 + 3x_2 + x_3 \leq 5$$
$$x_1 + 2x_2 + 2x_3 \leq 8$$
$$x_1, x_2, x_3 \geq 0$$

□

Problem 4 (20pts). Construct the KKT conditions for the following nonlinear program:

$$\begin{array}{rc} \text{minimize} & x_1 ln(x_1) + (x_2 - 2)^2 + x_3 \\ \text{subject to} & x_1 + x_2 \leq 3 \\ & x_3 - x_2^2 \geq 3 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

Solution:

Let λi be the Lagrangian multiplier of the ith constraint.
Lagrangian Function:

$$L(x, \lambda) = x_1 log(x_1) + (x_2 - 2)^2 + x_3 + \lambda_1(x_1 + x_2 - 3) + \lambda_2(x_3 - x_2^2 - 3)$$

Main condition:

$$log(x_1) + 1 + \lambda_1 \geq 0$$
$$2(x_2 - 2) + \lambda_1 - 2\lambda_2 x_2 \geq 0$$
$$1 + \lambda_2 \geq 0$$

Complementary Condition:

$$\lambda_1(x_1 + x_2 - 3) = 0$$
$$\lambda_2(x_3 - x_2^2 - 3) = 0$$
$$x_1(log(x_1) + 1 + \lambda_1) = 0$$
$$x_2(2(x_2 - 2) + \lambda_1 - 2\lambda_2 x_2) = 0$$
$$x_3(1 + \lambda_2) = 0$$

Dual Feasibility:

$$\lambda_1 \geq 0$$
$$\lambda_2 \leq 0$$

Primal Feasibility:

$$x_1 + x_2 \leq 3$$
$$x_3 - x_2^2 \geq 3$$
$$x_1, x_2, x_3 \geq 0$$

□

Problem 5 (20pts). FONC for Constrained Problems If x^* is a local minimum for continuous and differentiable function f, then, for any feasible direction d at x^*, we must have $\nabla f(x^*)^T d \geq 0$

Solution: If there is a feasible direction d at x^*, satisfied that $\nabla f(x^*)^T d < 0$, we denote

$$g(\alpha) = f(x^* + \alpha d)$$

and

$$g'(\alpha) = \nabla f(x^* + \alpha d)^T d$$

It implies that

$$g(0) = f(x^*), \quad g'(0) = \nabla f(x^*)^T d < 0$$

Thus, since $g(\cdot)$ is continuous, there exists $\delta > 0$, and for any $\alpha^0 \leq \delta$, we have $g(\alpha^0) < g(0)$, which implies that $f(x^* + \alpha^0 d) < f(x^*)$ for any $\alpha^0 \leq \delta$. Besides, with the fact that d is a feasible direction, there exists a feasible point x_1, s.t. $f(x_1) < f(x^*)$, which contradicts with the assumption that x^* is a local minimum. □